Microscopes II

Carefully read through the passage. Then select words from the list to fill in the spaces. Each word is only used once.

BOTANIST	CELLS	CENTURY	COMPOUND	DIVISION	DYES	ELECTRON
EYEPIECE	FOCUS	HOOKE	LENSES	LIGHT	MAGNIFIED	MICROSCOPE
MICROORGANISMS	MILLIONTHS	MOLECULES	PLANTS	OBJECTIVE	RESOLUTION	
SPECIMENS	SLIDES	SMALL	SMALLER	STAINED	THOUSANDS	VIRUSES

Microscopes give _____ images of objects that are difficult to see with unaided vision. Two main types of microscope are in common usage, the conventional _____ microscope that uses visible light and glass lenses and the _____ microscope that uses a beam of electrons. There are two types of electron microscope – the transmission _____ and the scanning electron microscope. All light microscopes nowadays are _____ microscopes, which have two sets of _____ arranged at opposite ends of a (body) tube. The _____ (ocular lens) is at the top of the tube and the _____ lens is at the bottom. The objects being looked at by microscopes are called _____ and are usually placed on glass _____. The distance from the tube to the specimen is altered to give an image that is in _____. Sometimes the parts of the specimen can't be seen clearly and need to be _____ using a dye. Light microscopes can only magnify a few _____ of times. The _____ of a light microscope is very important – this is a measure of how well a microscope can distinguish between very _____ objects that are close together. The light microscope can only give images of objects that around 500 nanometres, that is 500 _____ of a metre (10^{-9} m) The electron microscope can show items thousands of times _____ than this. Light microscopes can show the larger structures of cells and the shape of bacteria but cannot be used to observe _____. Electron microscopes can give images of viruses and even large _____ .

The microscope has been an essential tool for biologists in their study of organisms and _____. The first compound microscopes were invented at the end of the sixteenth _____ . The word 'cell' was first used by Robert _____ (1665). Antoni van Leeuwonhoek is credited with seeing the first living _____(1676). Robert Brown, a _____, discovered the nucleus in plant cells (1831). Matthias Schleiden stated that all _____ were made of cells (1837) and a year or so later his colleague, Theodor Schwann, came to a similar conclusion that cells were the basic unit of animal structure. Rudolf Virchow proposed that all cells arose from cell _____ (1859). Walther Flemming pioneered the use of _____ to stain cells. He was the first to observe mitosis (1880).

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